HARD · INFORMATION AND IDEAS

# Inferences

# 01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

**Q1** INFORMATION AND IDEAS

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because ______blank

Which choice most logically completes the text?

**A.** it can only be used on burrows below a certain size.

**B.** it does not allow for multiple castings of the same burrow over time.

**C.** the casting process takes more time than *A. bellulus* takes to construct a burrow.

**D.** the process of recovering the model distorts the resin's shape.

Q2 INFORMATION AND IDEAS

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues' recommendation therefore implies that the HST ______blank

Which choice most logically completes the text?

A. will likely continue to be used primarily to observe objects outside the solar system.

B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.

C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.

D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

The morphological novelty of echinoderms—marine invertebrates with radial symmetry, usually starlike, around a central point—impedes comparisons with most other animals, in which bilateral symmetry on an anterior-posterior (head to tail) axis through a trunk is typical. Particularly puzzling are sea stars, thought to have evolved a headless layout from a known bilateral origin. Applying genomic knowledge of *Saccoglossus kowalevskii* acorn worms (close relatives of sea stars, and thus expected to have similar markers for corresponding anatomical regions) to the body patterning genes of *Patiria miniata* sea stars, Laurent Formery et al. observed activity only in anterior genes across *P. miniata*'s entire body and some posterior genes limited to the edges, suggesting that ______blank

Which choice most logically completes the text?

**A.** despite the greater prevalence of anterior genes in sea stars' genetic makeup, posterior genes active at the body's perimeter are primarily responsible for the starlike layout that distinguishes sea stars' radial symmetry from that of other echinoderms.

**B.** contrary to the belief that they evolved from early ancestors with the bilateral form typical of many other animals, sea stars instead originated with an atypical body layout that was neither bilaterally nor radially symmetrical.

**C.** although the two species are closely related, there is only minimal correspondence in the genetic markers for head, tail, and trunk region development in *P. miniata* sea stars and *S. kowalevskii* acorn worms.

**D.** rather than undergoing changes resulting in the eventual elimination of a head region in their radial body plan, as previously assumed, sea stars' morphology evolved to completely lack a trunk and consist primarily of a head region.

**Q4** INFORMATION AND IDEAS

The increased integration of digital technologies throughout the process of book creation in the late 20th and early 21st centuries lowered the costs of book production, but those decreased costs have been most significant in the manufacturing and distribution process, which occurs after the authoring, editing, and design of the book are complete. This suggests that in the late 20th and early 21st centuries, ______blank

Which choice most logically completes the text?

**A.** digital technologies made it easier than it had been previously for authors to write very long works and get them published.

**B.** customers generally expected the cost of books to decline relative to the cost of other consumer goods.

**C.** publishers increased the variety of their offerings by printing more unique titles but also printed fewer copies of each title.

**D.** the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.

Henry Ossawa Tanner's 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore ______blank

Which choice most logically completes the text?

**A.** risk judging Tanner's painting by standards that may not be historically appropriate.

**B.** tend to conflate Tanner's political views with those of his contemporaries.

**C.** forgo analyzing Tanner's painting in favor of analyzing his political activity.

**D.** wrongly assume that Tanner's painting was intended as a critique of his fellow artists.

Like many other genera of wild bees, bumblebees have in recent decades experienced population collapse caused by, among other factors, habitat destruction and climate variation. Bumblebees are also one of the most researched bee genera, second only to honeybees. As a result, ecologists have gained much of their insight about wild-bee declines from bumblebees. In a 2021 paper, zoologist Guillaume Ghisbain notes that bumblebees are among the relatively few wild-bee genera that display social behaviors and dietary generalism (ability to obtain nectar and pollen from a diversity of plant species), two traits that are associated with increased resilience to some specific environmental changes. Ghisbain therefore contends that ______blank

Which choice most logically completes the text?

**A.** although bumblebees and many other wild bees have experienced similar population declines in the past, compared with other wild bees, bumblebees are likely at greater risk of being harmed by climate variation than by habitat destruction.

**B.** although bumblebees have been more extensively studied than most wild bees, researchers should not use bumblebees to draw conclusions about the decline of other wild bees, even ones with feeding patterns and levels of sociability that are similar to those of bumblebees.

**C.** because bumblebees and other bees with generalist diets are less negatively affected by environmental stress than bees with specialized diets are, they are less likely to experience major population changes in the future than bees with specialized diets are.

**D.** because the responses of bumblebees and other wild bees to environmental threats are not always comparable, researchers need to exercise caution when extrapolating information about wild-bee population declines from bumblebees.

Q7



A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that ______blank

Which choice most logically completes the text?

**A.** temperature-sensitive accelerated flowering is unique to *A. thaliana*.

**B.** *A. thaliana* increases ELF3 production as temperatures rise.

**C.** ELF3 enables *A. thaliana* to respond to increased temperatures.

**D.** temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylanderi* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host's fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylanderi* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the ______blank

Which choice most logically completes the text?

**A.** uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected workers in the nest.

**B.** need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

**C.** high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly disperse when the nest is attacked by a predator.

**D.** average lifespan of *T. nylanderi* worker ants in colonies without parasitic activity typically falls well below three years, the range covered by the study.

Q9

INFORMATION AND IDEAS

An analysis by Alain Elayi and colleagues of coins minted in Sidon in the fifth and fourth centuries BCE reveals a change in their composition over time: while a coin from circa 450 BCE contains about 98% silver and 1% copper, a coin from 367 BCE (the end of Ba'alšillem II's reign) contains 74.2% silver and 24.7% copper, giving it a relatively yellowish appearance that traders would have noticed. Because coins with a silver content below 80% were widely considered unsuitable for trade, Elayi et al. speculate that a crisis in confidence in the currency occurred in Sidon around 367 BCE, which was likely relieved—despite Sidon's persistent oppressive financial obligations—as a result of Ba'alšillem II's successor Abd'aštart I's decision to ______blank

Which choice most logically completes the text?

**A.** proclaim that the percentage of silver in coins suitable for trade would be raised to a threshold higher than 80%.

**B.** keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but decrease their weight.

**C.** begin minting heavier coins with a proportion of silver to copper similar to that in coins minted in 367 BCE.

**D.** fund the mining of some copper deposits that were not available to Ba'alšillem II.

Q10 INFORMATION AND IDEAS

During the Bourbon Restoration in France (1814–1830), the right to vote required in part that a person paid at least 300 francs in direct taxes to the government. The four most common taxes (the *quatre vieilles*) were levied on real estate (both land and buildings); the doors and windows in taxpayer homes; the rental values of homes; and the businesses of artisans and merchants. (Foreign investments were either exempt from taxation or taxed lightly.) Although relatively few people paid the tax on real estate, it was the main means of voter qualification and accounted for over two-thirds of government receipts during this period, suggesting that during the Bourbon Restoration ______blank

Which choice most logically completes the text?

A. those people who had the right to vote most likely had substantial holdings of French real estate.

B. the voting habits of French artisans and merchants were effective in reducing tax burdens on businesses.

C. the number of doors and windows in French residences was kept to a minimum but increased after 1830.

D. French people with significant foreign investments were unlikely to have the right to vote.